

Political Bias and Temporal Dynamics in Large Language Models

Anonymous ACL submission

Abstract

LLMs exhibit systematic political biases, but most work treats bias as static. We study political bias as a temporal phenomenon using Prediction Inconsistency (IC) and sentiment label distributions. We evaluate ChatGPT (gpt-3.5-turbo through gpt-4o) and Llama 3.2 (Base and Instruct) on target-oriented sentiment classification. IC jumps from ≈ 0 (gpt-3.5-turbo) to ≈ 0.67 (gpt-4) and plateaus at ≈ 0.61 for later GPT-4 variants; Llama IC is lower (0.005–0.178) and instruction tuning strongly reduces it for the 1B model. Label distributions shift from all-positive (gpt-3.5-turbo) to positive/UNKNOWN mixes (GPT-4) and vary by size and alignment for Llama.

1 Introduction

LLMs are increasingly used to access and interpret political information; their latent political orientations have real consequences (????). Most prior work evaluates bias at a single point in time, yet model families evolve rapidly in training data, safety, and alignment (??). We study political bias over time using Prediction Inconsistency (IC) (?) and sentiment label distributions on two families: OpenAI ChatGPT and Meta Llama. We ask: (1) How does IC change across model generations? (2) How does alignment (Base vs. Instruct) affect IC and label distribution? Our contributions: a temporal view of political bias, longitudinal IC and label-distribution results for ChatGPT and Llama, and evidence that alignment effects are size- and family-dependent.

2 Related Work

Prior work measures political bias via ideological mapping (??), stance vs. framing (?), and target-oriented sentiment (??). Temporal drift in LLM behavior and bias is documented (??); LLaMA evolution illustrates changes in safety and alignment

(??). Mitigation and alignment can alter political nuance (??). We extend ? to a temporal, multi-family setting.

3 Methodology

We follow ? : 250 political entities, 60 sentence templates (positive/neutral/negative), entity substitution, and target-oriented sentiment classification (TSC) with labels {positive, neutral, negative}. Political bias is Prediction Inconsistency (IC): entropy of predicted labels across entities per template, averaged over templates. For template s , $P(l|s) = |\{e \in E : \text{pred}(s, e) = l\}|/|E|$; $H(s) = -\sum_l P(l|s) \log P(l|s)$; $\text{IC} = \frac{1}{m} \sum_i H(s_i)$. $\text{IC}=0$ means perfectly consistent (unbiased). We use 9-shot prompting, temperature=0, and identical prompt sets across versions. Models: ChatGPT (gpt-3.5-turbo, gpt-4, gpt-4-turbo, gpt-4o-mini, gpt-4o); Llama 3.2 Base (1b, 3b) and Instruct (1b, 3b, vision 11b).

4 Results

ChatGPT (Figure 1, left). IC is near zero for gpt-3.5-turbo, jumps to ≈ 0.67 for gpt-4, and plateaus at ≈ 0.61 for gpt-4-turbo, gpt-4o-mini, and gpt-4o (mean 0.501). Label distribution: gpt-3.5-turbo 100% positive; gpt-4 $\approx 60\%$ positive, 40% UNKNOWN; gpt-4-turbo / gpt-4o-mini / gpt-4o $\approx 30\%$ positive, 70% UNKNOWN. No neutral or negative in this eval.

Llama (Figure 1, right). Base: IC 1b ≈ 0.178 , 3b ≈ 0.118 . Instruct: 1b ≈ 0.028 , 3b ≈ 0.120 , vision 11b ≈ 0.005 . Instruction tuning strongly reduces IC for 1b; 3b is similar in Base and Instruct; 11b has lowest IC. Label distribution: Base 1b/3b skew negative; Instruct 1b $\approx 97\text{--}98\%$ negative; Instruct 3b and 11b show more positive/neutral (e.g., 11b $\approx 32\%$ positive, 30% neutral, 34% negative).

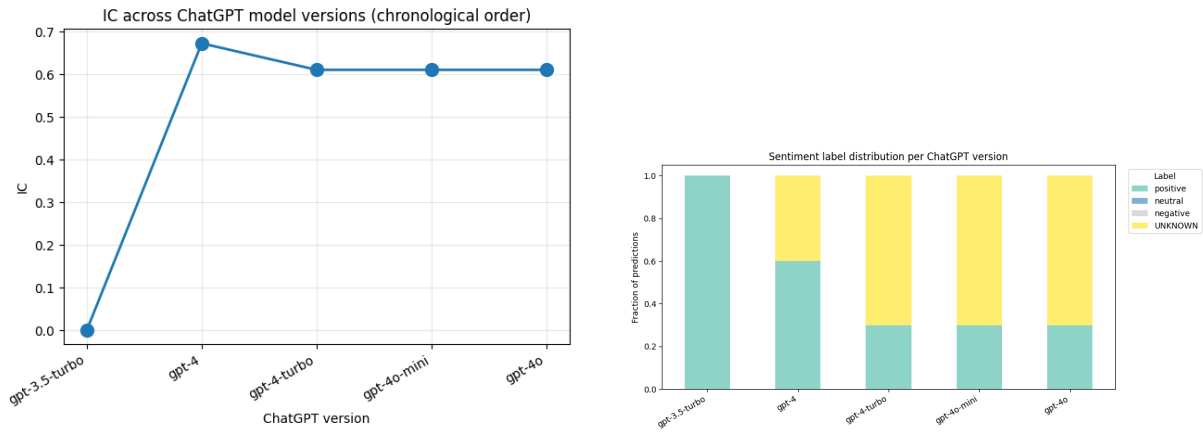


Figure 1: Left: IC across ChatGPT versions (chronological). Right: IC for Llama 3.2 Base vs. Instruct.

5 Discussion & Conclusion

ChatGPT shows a sharp IC increase from gpt-3.5-turbo to gpt-4 and a plateau for later gpt-4 variants; the shift to large UNKNOWN shares may reflect refusal/hedging while still yielding high inconsistency. For Llama, instruction tuning reduces IC for 1B but not 3B; the 11B vision model has minimal IC. Alignment effects on both IC and label distribution are size- and family-dependent. Cross-family, ChatGPT has higher IC (0.61–0.67) than Llama (0.005–0.178) and different label patterns (positive/UNKNOWN vs. negative/neutral). Limitations: two families and fixed versions; IC does not capture all bias dimensions; UNKNOWN share complicates interpretation. Future work: more families, time horizons, and bias metrics.

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